

Cybersecurity Enhancement Through Malicious URL Detection With ML

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Introduction

In today's digital world, people use the internet for shopping, banking, and social media, but some websites are dangerous and can steal personal information or harm devices. Cybercriminals create phishing and malware sites, making it important to detect threats quickly. Traditional methods like blacklists are often slow and outdated. To address this, we developed a machine learning-based system that analyzes website URLs, content, domain age, SSL certificates, and more to detect malicious sites. Using models like Random Forest and SVM, our tool predicts if a website is safe, suspicious, or harmful. It also connects with trusted sources like Google Safe Browsing and WHOIS for added accuracy. The system continuously learns from new data, improving over time to stay effective against emerging threats. Our real-time tool helps users check website safety and offers confidence scores. Future plans include browser extensions and mobile apps for enhanced protection. We created a machine learning system to detect harmful websites by analyzing URLs, domain info, and security factors. It uses models like Random Forest and SVM, along with data from sources like Google Safe Browsing, to predict if a site is safe. The system continuously updates to protect users in real time, with plans for browser and mobile integration.

Proposed System

Detecting malicious URLs is vital to protect users from phishing, malware, and scams. These harmful links often look normal but can steal information or install viruses. A structured process is needed to build an effective detection system: data collection, feature extraction, algorithm selection, model training, and evaluation. First, a diverse dataset of safe and malicious URLs is gathered from trusted sources. Then, key features like URL length, special characters, subdomains, and suspicious keywords are extracted to help the model spot threats. Machine learning algorithms like Naive Bayes, Decision Trees, Random Forest, and KNN are tested to find the most accurate one. The selected model is trained on labeled data and evaluated using precision, recall, and F1-score to ensure it accurately detects malicious URLs while minimizing false positives. Continuous updates with new data and techniques are essential to keep the system effective against evolving cyber threats.



Existing System

Current methods for detecting malicious URLs use machine learning and lexical analysis but have limitations in real-time accuracy.

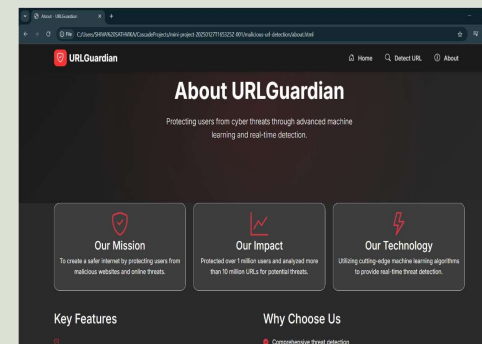
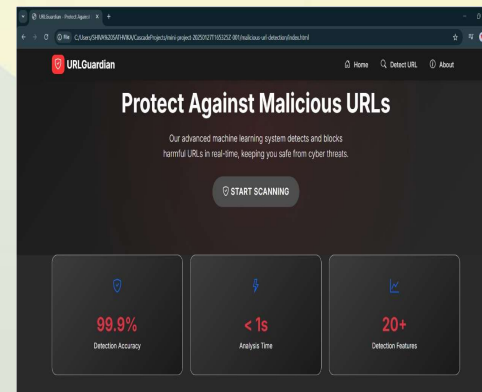
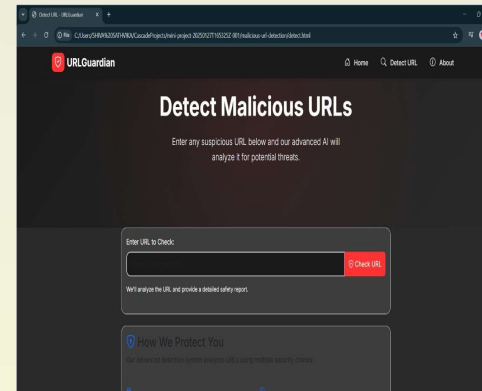
1. Phishing Detection with Machine Learning & Lexical Features: Analyzes URL structure (length, symbols, keywords) to detect threats. Limitation: Struggles with new phishing URLs that don't match known patterns.

Suspicious URL Filtering with Logistic Regression & Multi-2.View Analysis: Examines domains, paths, and historical data from multiple perspectives. Limitation: Less effective with complex data and evolving phishing techniques.

3. Malicious URL Detection with Combined SVM & Logistic Regression: Merges SVM's feature handling with Logistic Regression's probability estimates. Limitation: Challenges in real-time detection and external data validation reduce efficiency.



Results



Conclusions

In an era where cyber threats are growing increasingly sophisticated, an automated system for detecting malicious URLs is essential for enhancing online security. Our project leverages machine learning techniques to analyze various URL characteristics and predict whether a website is safe or harmful. By incorporating features such as URL structure, domain reputation, SSL validity, and content patterns, our model can effectively identify phishing sites, malware-hosting pages, and other cyber threats. Through the use of machine learning algorithms like Random Forest, Support Vector Machines, and Decision Trees, our system continuously learns from new data to improve detection accuracy. The integration of real-time tools, external security databases, and continuous updates ensures that users receive the most reliable and up-to-date protection against emerging threats. While our current system provides strong detection capabilities, future enhancements may include browser extensions, mobile applications, and AI-driven adaptive learning techniques. These improvements will further strengthen cybersecurity measures and help individuals and organizations stay safe in the ever-evolving digital landscape. By continuously refining our approach and expanding our capabilities, we aim to make the internet a safer place for everyone.

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